EMERGENCY MEDICINE ROTATION EXAM

CHAPTER 4: PULMONOLOGY — HIGH-YIELD REVIEW

1. ACUTE BRONCHIOLITIS

PATHOPHYSIOLOGY & ETIOLOGY

Viral infection of the lower respiratory tract primarily caused by **Respiratory Syncytial Virus (RSV)** (>70% of cases). Characterized by inflammatory obstruction of small airways (bronchioles) in infants and young children (<2 years old). Peak incidence occurs during winter and early spring months.

CLINICAL PRESENTATION

Viral prodrome (fever, coryza, nasal congestion) lasting 1-3 days, escalating to acute respiratory distress with tachypnea, expiratory wheezing, intercostal/subcostal retractions, nasal flaring, and poor feeding.

PHYSICAL EXAMINATION

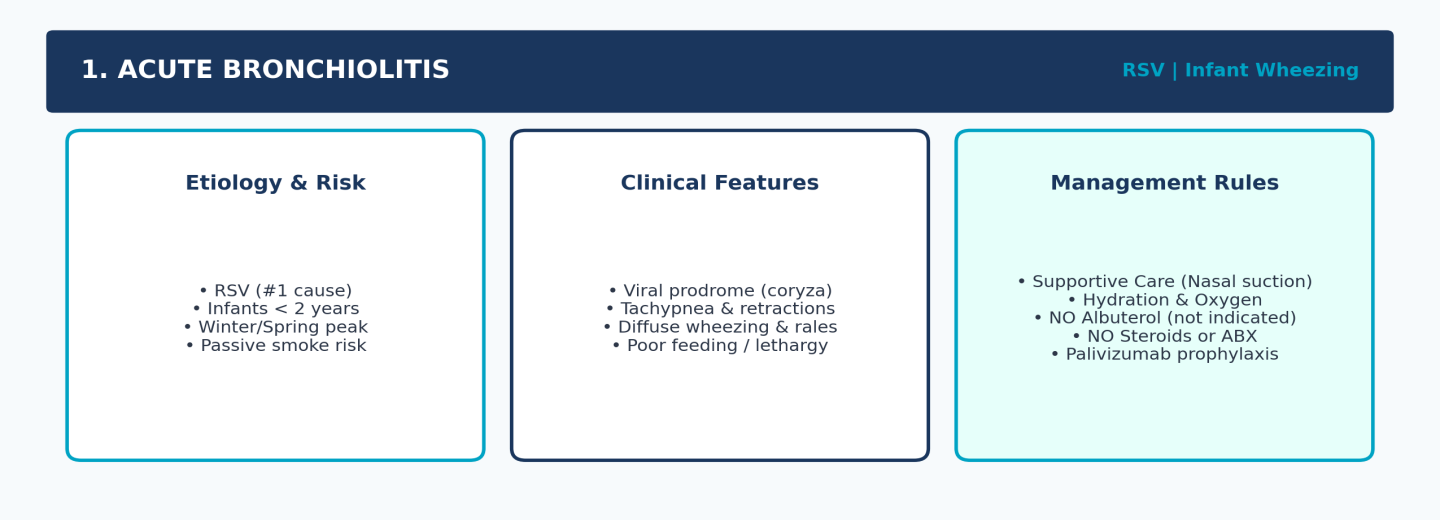
Tachypnea, prolonged expiratory phase, diffuse expiratory wheezing, scattered fine rales/crackles, hyperresonance on percussion, and variable hypoxia (SpO2 < 92%).

DIAGNOSTIC STUDIES

Clinical diagnosis based on age and examination. Nasopharyngeal PCR/antigen swab confirms RSV. Chest X-ray (indicated only in severe or atypical cases) shows bilateral hyperinflation, peribronchial cuffing, and patchy atelectasis.

MANAGEMENT & TREATMENT

Supportive care is the mainstay: Nasal suctioning, humidified oxygen, and adequate hydration. **Contraindicated interventions:** Routine bronchodilators (Albuterol), systemic corticosteroids, or antibiotics are NOT recommended. *Palivizumab* monoclonal antibody prophylaxis for high-risk premature infants.



HIGH-YIELD BOARD PEARL: BOARD PEARL: Bronchiolitis is the #1 cause of hospitalization in infants < 1 year. Do NOT order Albuterol, Steroids, or Antibiotics on board exams — treatment is strictly supportive with airway suctioning and hydration!

2. ACUTE BRONCHITIS

PATHOPHYSIOLOGY & ETIOLOGY

Self-limiting inflammation of the large airways (trachea and bronchi). **Over 90% of cases are VIRAL** (Adenovirus, Influenza A/B, Parainfluenza, RSV, Rhinovirus). Bacterial causes (M. pneumoniae, C. pneumoniae, B. pertussis) account for <10%.

CLINICAL PRESENTATION

Cough lasting 1 to 3 weeks (hallmark feature). Cough is initially dry, then often becomes productive with clear or purulent sputum (purulence reflects inflammatory desquamation, NOT bacterial superinfection!). Associated with mild low-grade fever, malaise, and retrosternal discomfort.

PHYSICAL EXAMINATION

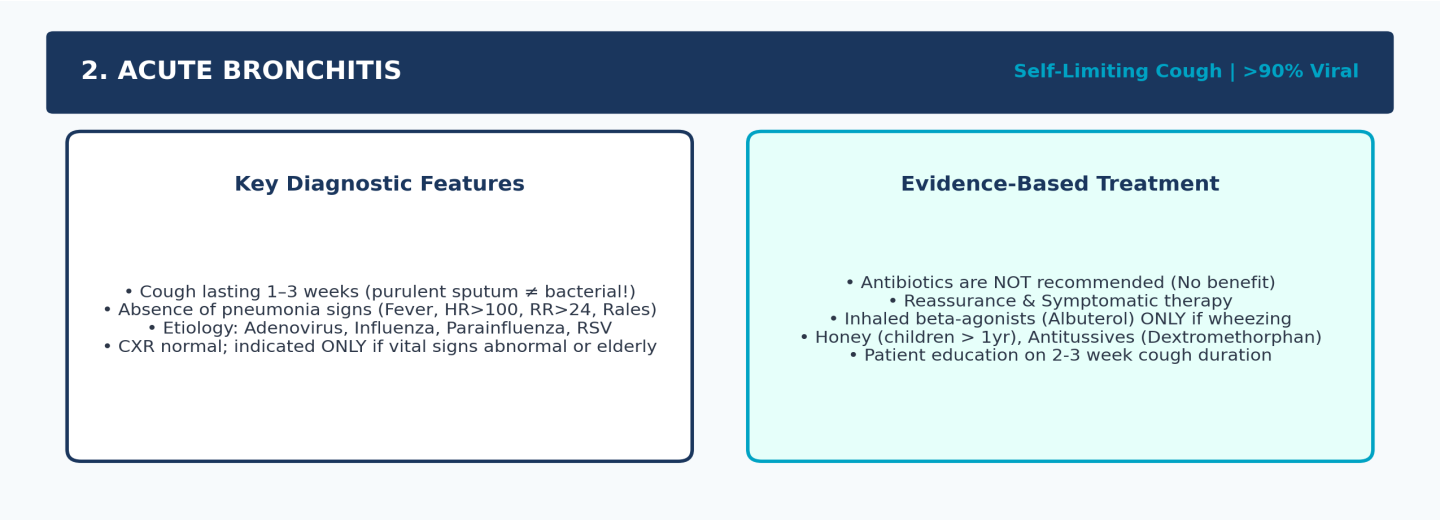
Lung fields are typically clear, or exhibit scattered rhonchi and expiratory wheezing that **clears or shifts with coughing**. Absence of signs of consolidation or systemic shock.

DIAGNOSTIC STUDIES

Clinical diagnosis. Chest X-ray is normal and indicated ONLY if pneumonia is suspected (e.g., HR > 100, RR > 24, Temp > 38.0°C, or focal rales/consolidation on exam). Procalcitonin testing can reduce unneeded antibiotic prescriptions.

MANAGEMENT & TREATMENT

Antibiotics are NOT recommended for acute bronchitis in immunocompetent patients. Treatment is purely symptomatic: NSAIDs/Acetaminophen for discomfort, Antitussives (Dextromethorphan/Codeine), and inhaled SABA (Albuterol) ONLY if wheezing is present. Patient education regarding expected 2-3 week cough duration.



HIGH-YIELD BOARD PEARL: BOARD PEARL: Purulent sputum DOES NOT equal bacterial bronchitis! Routine antibiotic therapy provides zero benefit and increases drug resistance. Educate patients that cough normally lasts 10 to 21 days.

3. ACUTE EPIGLOTTITIS

PATHOPHYSIOLOGY & ETIOLOGY

Life-threatening bacterial inflammation and edema of the epiglottis and supraglottic structures leading to rapid, fatal upper airway obstruction. Historically caused by *Haemophilus influenzae* type b (Hib); now more common in adults and unimmunized children due to Group A Strep, S. pneumoniae, and S. aureus.

CLINICAL PRESENTATION

Abrupt onset of severe sore throat, high fever, dysphagia, odynophagia, and muffled 'hot potato' voice. Patients assume the **'Tripod Position'** (leaning forward, neck extended, chin thrust out).

PHYSICAL EXAMINATION

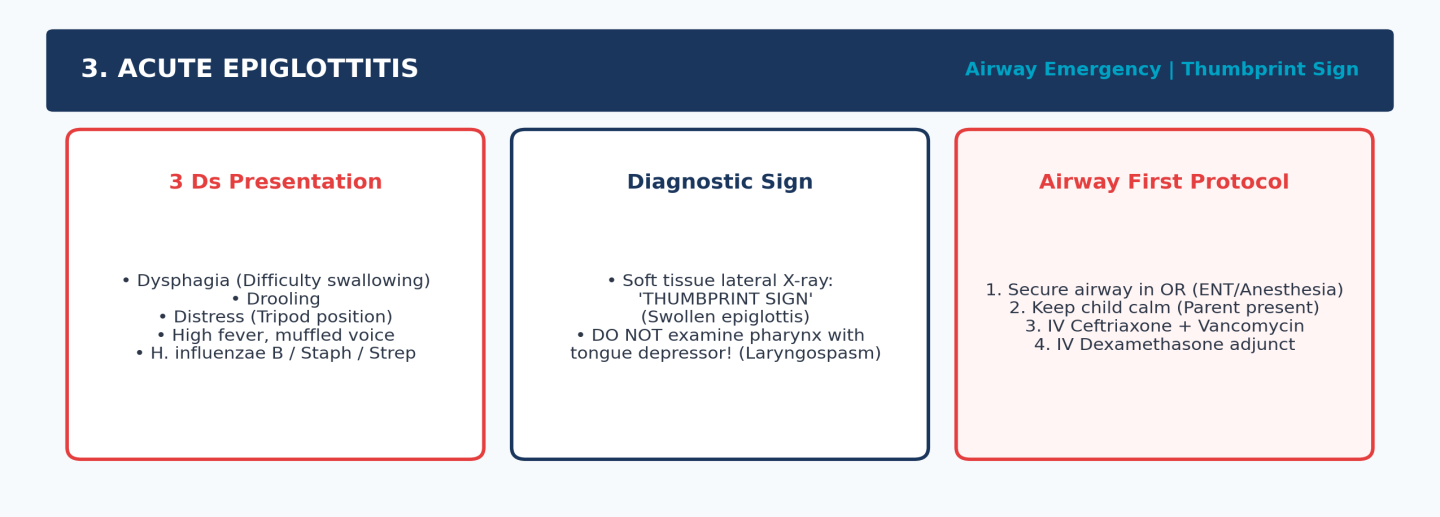
Classic 3 Ds: Dysphagia, Drooling, and Distress. Inspiratory stridor, tachypnea, and toxic appearance. **CONTRAINDICATION:** NEVER inspect the pharynx with a tongue depressor or anxiety-inducing procedure in children — may precipitate lethal laryngospasm!

DIAGNOSTIC STUDIES

Lateral soft-tissue neck Radiograph reveals the classic **'THUMBPRINT SIGN'** (swollen, enlarged epiglottis). Definitive diagnosis via direct laryngoscopy in a controlled setting (OR).

MANAGEMENT & TREATMENT

Immediate Airway Protection is Top Priority: Secure airway in the Operating Room with Anesthesia/ENT present. Keep patient calm. Medical therapy: IV Ceftriaxone/Cefotaxime + Vancomycin (for MRSA) plus IV Dexamethasone to reduce airway edema.



HIGH-YIELD BOARD PEARL: BOARD PEARL: Child with drooling, tripod stance, and stridor = Epiglottitis! Keep child in parent's lap, do NOT insert tongue blade or agitate, and prepare for emergent airway management in the OR.

4. ACUTE RESPIRATORY DISTRESS SYNDROME (ARDS)

PATHOPHYSIOLOGY & ETIOLOGY

Life-threatening inflammatory lung injury secondary to systemic insults (Sepsis #1 cause, Trauma, Aspiration, Pancreatitis, Pneumonia). Inflammatory cascade leads to diffuse alveolar damage (DAD), capillary endothelial leakage, surfactant inactivation, and severe non-cardiogenic pulmonary edema.

CLINICAL PRESENTATION

Rapid onset (within 24-48 hours of triggering insult) of severe dyspnea, tachypnea, anxiety, and refractory hypoxemia (hypoxia unresponsive to high-flow oxygen).

PHYSICAL EXAMINATION

Severe respiratory distress, intercostal retractions, cyanosis, diffuse bilateral rales/crackles. Absence of clinical signs of congestive heart failure (no JVD, S3, or peripheral edema).

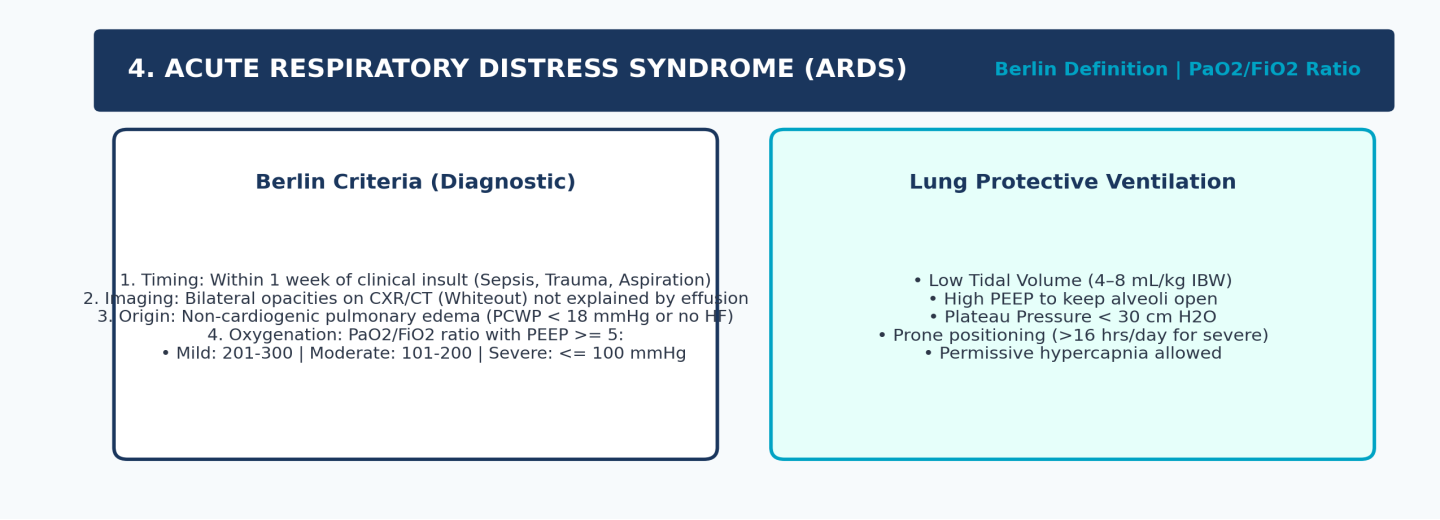
DIAGNOSTIC STUDIES

Berlin Definition Criteria:

1. Timing within 1 week of clinical insult.
2. Bilateral opacities on CXR/CT ('Whiteout') not fully explained by effusions/atelectasis.
3. Non-cardiogenic origin (PaOP / PCWP < 18 mmHg or normal Echo).
4. Oxygenation deficit: **PaO2/FiO2 ratio <= 300 mmHg with PEEP >= 5 cmH2O** (Mild: 201-300, Moderate: 101-200, Severe: <= 100).

MANAGEMENT & TREATMENT

Lung Protective Mechanical Ventilation: Low Tidal Volume strategy (4-8 mL/kg IBW) to prevent volutrauma, High PEEP to recruit collapsed alveoli, and Plateau Pressure limit < 30 cmH2O. Prone positioning for severe ARDS (PaO2/FiO2 < 150). Conservative fluid management and treatment of underlying cause.



HIGH-YIELD BOARD PEARL: BOARD PEARL: ARDS presents with severe hypoxia refractory to 100% O2 with bilateral infiltrates and normal PCWP (<18). Key ventilator management is LOW TIDAL VOLUME (4-8 mL/kg) and HIGH PEEP!

5. ACUTE ASTHMA EXACERBATION

PATHOPHYSIOLOGY & ETIOLOGY

Chronic inflammatory airway disease characterized by reversible airway hyperresponsiveness, bronchospasm, mucosal edema, and mucus plugging. Exacerbations triggered by viral infections, allergens, exercise, cold air, or medication non-compliance.

CLINICAL PRESENTATION

Acute onset of dyspnea, chest tightness, wheezing, and dry cough. Patients may speak in short phrases or monosyllables depending on severity.

PHYSICAL EXAMINATION

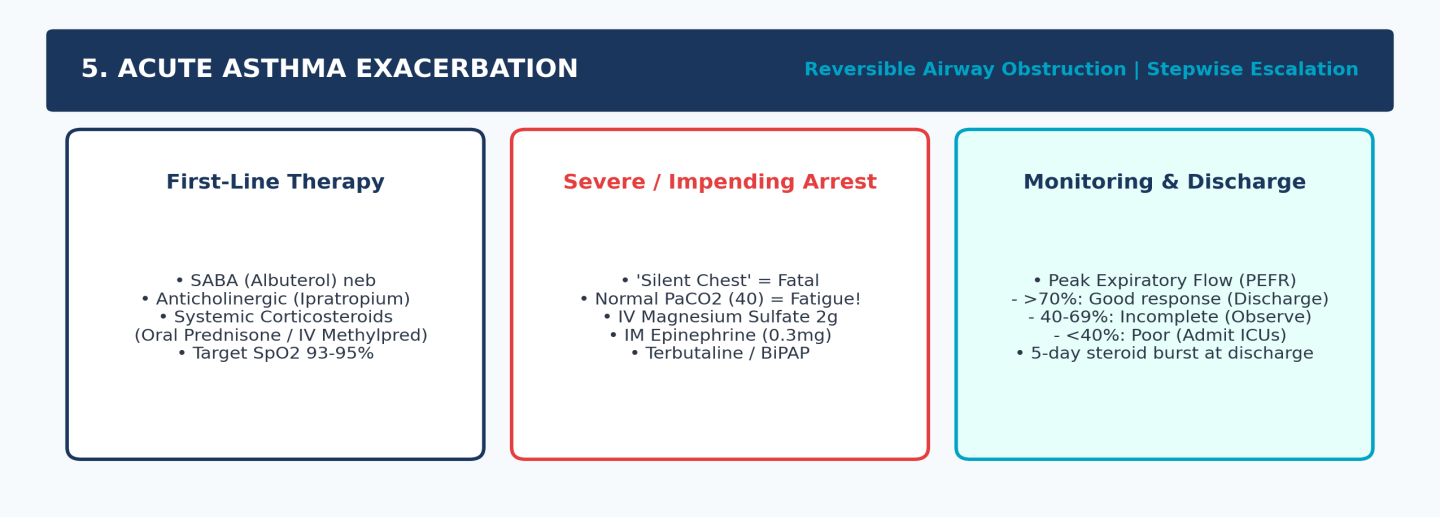
Tachypnea, prolonged expiratory phase, diffuse high-pitched expiratory wheezing, accessory muscle use. **RED FLAG:** 'Silent Chest' (absence of wheezing/breath sounds) indicates severe impending respiratory arrest!

DIAGNOSTIC STUDIES

Peak Expiratory Flow Rate (PEFR): < 50% predicted indicates severe exacerbation. ABG initially shows respiratory alkalosis (pCO2 < 35 due to hyperventilation); **a normal or rising pCO2 (>= 40) signals respiratory fatigue and impending failure!**

MANAGEMENT & TREATMENT

First-line: Inhaled SABA (Albuterol) + SAMA (Ipratropium bromide) continuous nebulization + Early Systemic Corticosteroids (Oral Prednisone or IV Methylprednisolone). **Severe/Refractory:** IV Magnesium Sulfate (2g IV over 20 min), IM Epinephrine (0.3 mg), BiPAP, and Intubation as last resort.



HIGH-YIELD BOARD PEARL: BOARD PEARL: An asthmatic in severe distress with a 'normal' PaCO2 (40 mmHg) is NOT improving — they are tiring out and entering fatal hypercapnic failure! Prepare for aggressive escalation.

6. CROUP (LARYNGOTRACHEOBRONCHITIS)

PATHOPHYSIOLOGY & ETIOLOGY

Viral infection of the upper airway causing subglottic tracheal edema and inflammation. Most common in children 6 months to 3 years old. Caused primarily by **Parainfluenza Virus Type 1** (>75% of cases).

CLINICAL PRESENTATION

Viral URI prodrome (rhinorrhea, low fever) followed by sudden onset of a characteristic **'Barking' seal-like cough**, inspiratory stridor, and hoarseness, typically worsening at night.

PHYSICAL EXAMINATION

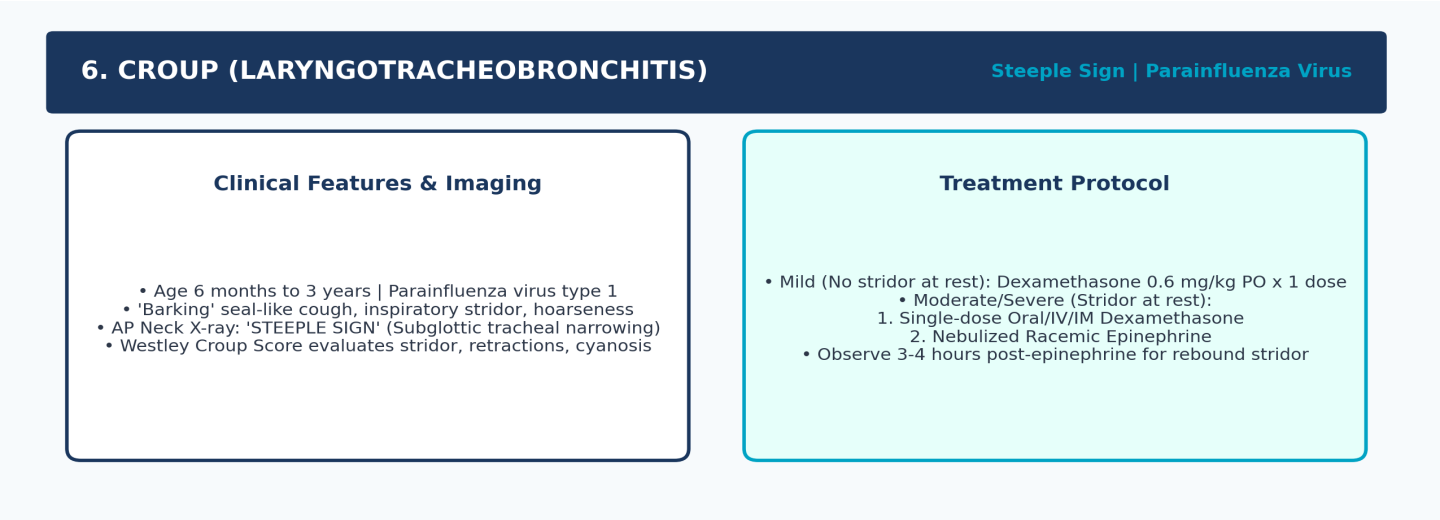
Inspiratory stridor, tachypnea, intercostal retractions. Westley Croup Score assesses severity (Stridor, Retractions, Air entry, Cyanosis, Level of consciousness).

DIAGNOSTIC STUDIES

Clinical diagnosis. Anteroposterior (AP) neck X-ray exhibits the classic **'STEEPLE SIGN'** (subglottic narrowing of the trachea).

MANAGEMENT & TREATMENT

Mild (No stridor at rest): Single dose of Dexamethasone (0.6 mg/kg PO/IM x 1).
Moderate to Severe (Stridor at rest): Dexamethasone PLUS **Nebulized Racemic Epinephrine**. Observe in ED for 3-4 hours post-epinephrine for rebound stridor before discharge.



HIGH-YIELD BOARD PEARL: BOARD PEARL: Child with barking seal cough + Steeple Sign on AP neck X-ray = Croup. Treat stridor at rest with Nebulized Racemic Epinephrine + Dexamethasone and observe for rebound stridor.

7. FOREIGN BODY ASPIRATION

PATHOPHYSIOLOGY & ETIOLOGY

Accidental inhalation of a foreign object (peanuts, organic seeds, small toys, coins) into the tracheobronchial tree. Peak incidence in toddlers (ages 1-3 years). Most commonly lodges in the **Right Mainstem Bronchus** due to its wider, shorter, and more vertical anatomical angle.

CLINICAL PRESENTATION

Sudden onset of severe choking, coughing, gagging, and acute respiratory distress. May be followed by an asymptomatic phase if the object settles, leading to delayed recurrent pneumonia or atelectasis.

PHYSICAL EXAMINATION

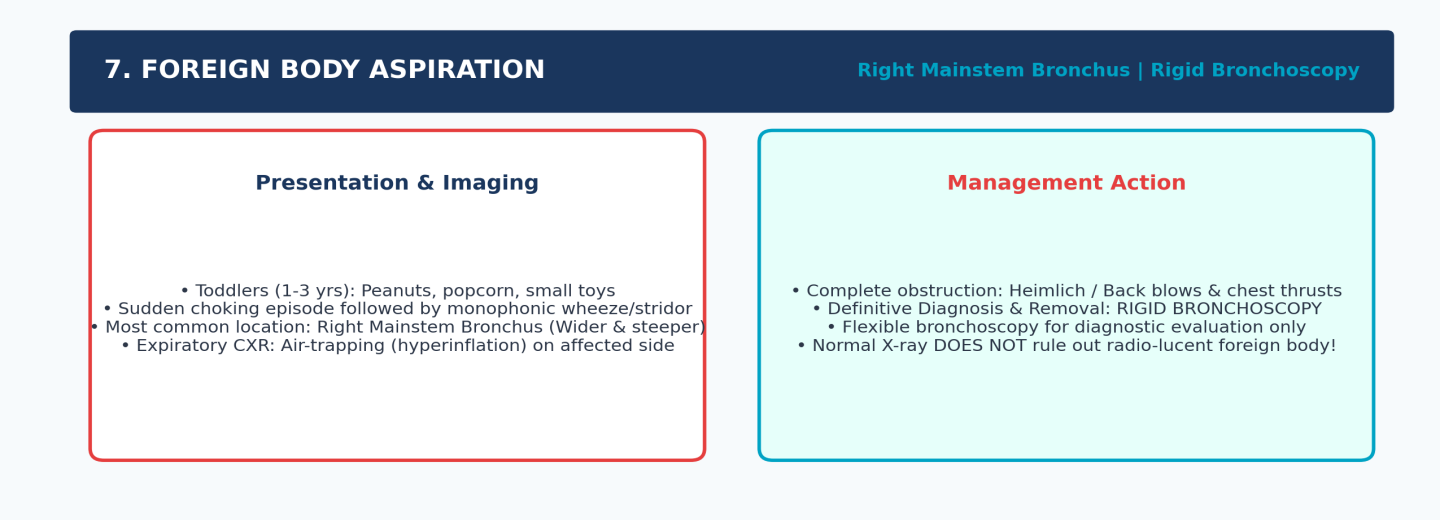
Unilateral monophonic wheezing, localized decreased breath sounds, tracheal deviation, or stridor (if upper airway).

DIAGNOSTIC STUDIES

Expiratory or Lateral Decubitus CXR demonstrates **air-trapping (hyperinflation) on the affected side**, atelectasis, or mediastinal shift away from the involved lung. Note: Radio-lucent objects (peanuts) will NOT appear on X-ray!

MANAGEMENT & TREATMENT

Complete obstruction: Heimlich maneuver (adults/children > 1yr) or 5 back blows / 5 chest thrusts (infants < 1yr). **Definitive Diagnosis & Management: RIGID BRONCHOSCOPY** under general anesthesia (allows airway control and object extraction).



HIGH-YIELD BOARD PEARL: BOARD PEARL: Unilateral localized wheezing in a toddler with sudden onset cough = Foreign Body Aspiration (Right Mainstem Bronchus #1). Rigid Bronchoscopy is required for therapeutic extraction!

8. HEMOPTYSIS

PATHOPHYSIOLOGY & ETIOLOGY

Expectoration of blood originating from the lower respiratory tract. **Massive Hemoptysis** defined as >100-600 mL blood in 24 hours (immediate asphyxiation threat). 90% of massive hemoptysis originates from the **Bronchial Arteries** (high-pressure systemic circulation), whereas non-massive hemoptysis often stems from pulmonary capillaries.

CLINICAL PRESENTATION

Coughing up bright red frothy blood or blood-streaked sputum. Differentiate from hematemesis (dark coffee-ground emesis, acidic pH) and pseudo-hemoptysis (upper airway bleeding).

PHYSICAL EXAMINATION

Signs of respiratory failure, focal rales, rhonchi, or bronchial breath sounds. Hemodynamic instability in massive cases.

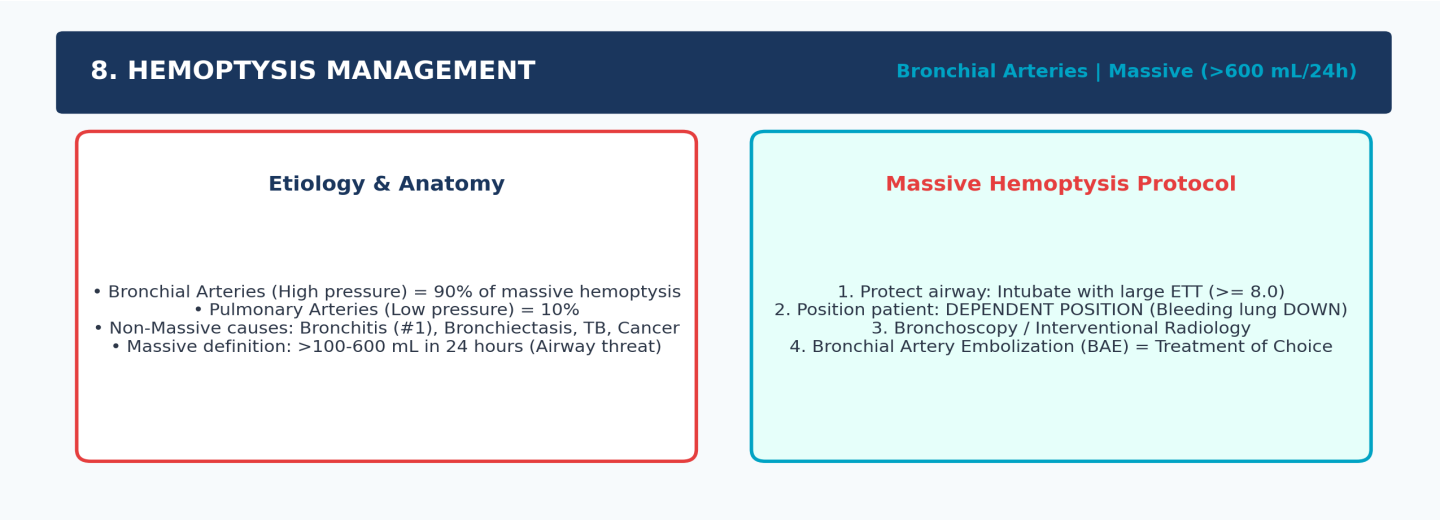
DIAGNOSTIC STUDIES

Chest X-ray to localize bleeding side. CT Angiography (CTA) of the chest is gold standard for identifying vascular etiology. Flexible Bronchoscopy to visualize site.

MANAGEMENT & TREATMENT

Massive Hemoptysis Emergency Protocol:

1. Secure airway with large ETT (>= 8.0 mm).
2. Position patient in **DEPENDENT POSITION (Bleeding lung DOWN)** to prevent blood from flooding the healthy lung.
3. Interventional Radiology: **Bronchial Artery Embolization (BAE)** is treatment of choice.



HIGH-YIELD BOARD PEARL: BOARD PEARL: In massive hemoptysis, asphyxiation (not exsanguination) is the cause of death. Immediately place the BLEEDING LUNG DOWN to preserve ventilation in the healthy lung!

9. INFLUENZA

PATHOPHYSIOLOGY & ETIOLOGY

Acute viral respiratory illness caused by **Influenza A or B viruses** (Orthomyxovirus family). Transmitted via respiratory droplets. Characterized by surface antigens Hemagglutinin (H) and Neuraminidase (N), subject to antigenic drift (seasonal epidemics) and antigenic shift (pandemics).

CLINICAL PRESENTATION

Abrupt onset of high fever, chills, prominent myagias, severe headache, malaise, non-productive cough, and sore throat.

PHYSICAL EXAMINATION

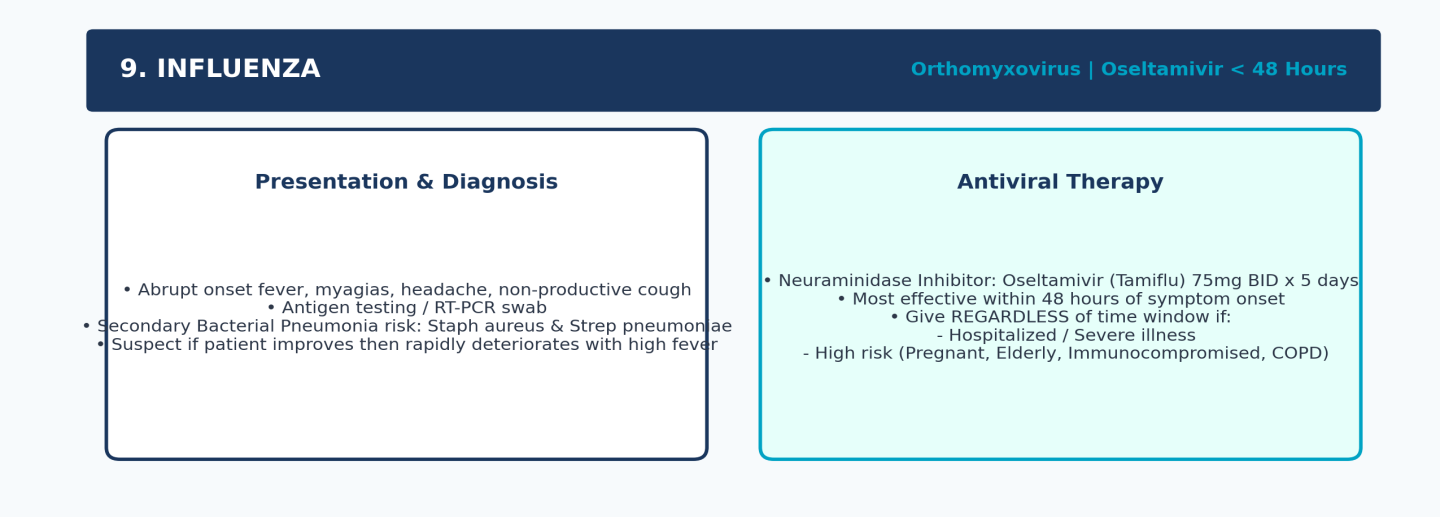
Fever, flushed face, conjunctival injection, pharyngeal erythema, and clear lung examination unless secondary bacterial pneumonia develops.

DIAGNOSTIC STUDIES

Rapid Influenza Molecular Assay (RT-PCR swab) or Rapid Antigen Test. Secondary Bacterial Pneumonia suspected if patient improves then rapidly deteriorates with high fever and purulent sputum (commonly *Staphylococcus aureus* or *Strep pneumoniae*).

MANAGEMENT & TREATMENT

Supportive care. **Oseltamivir (Tamiflu)** 75 mg PO BID x 5 days (Neuraminidase inhibitor). Most effective when initiated within **48 hours of symptom onset**. Give regardless of time window if patient is hospitalized, severely ill, or high-risk (pregnant, elderly, COPD, immunocompromised).



HIGH-YIELD BOARD PEARL: BOARD PEARL: Post-influenza pneumonia with rapid clinical deterioration and cavitary lung lesions is classic for secondary STAPHYLOCOCCUS AUREUS pneumonia! Treat with IV Vancomycin.

10. LUNG CANCER & PARANEOPLASTIC SYNDROMES

PATHOPHYSIOLOGY & ETIOLOGY

Leading cause of cancer mortality worldwide. Classified into **Non-Small Cell Lung Cancer (NSCLC - 85%)** and **Small Cell Lung Cancer (SCLC - 15%)**. Strongly linked to cigarette smoking (except Adenocarcinoma).

CLINICAL PRESENTATION

New or changing chronic cough, hemoptysis, unexplained weight loss, dyspnea, chest pain, recurrent pneumonia.

PHYSICAL EXAMINATION

Signs of consolidation, pleural effusion, supraclavicular lymphadenopathy, or specific tumor syndromes (e.g., SVC Syndrome: facial swelling, jugular vein distension, upper extremity edema).

DIAGNOSTIC STUDIES

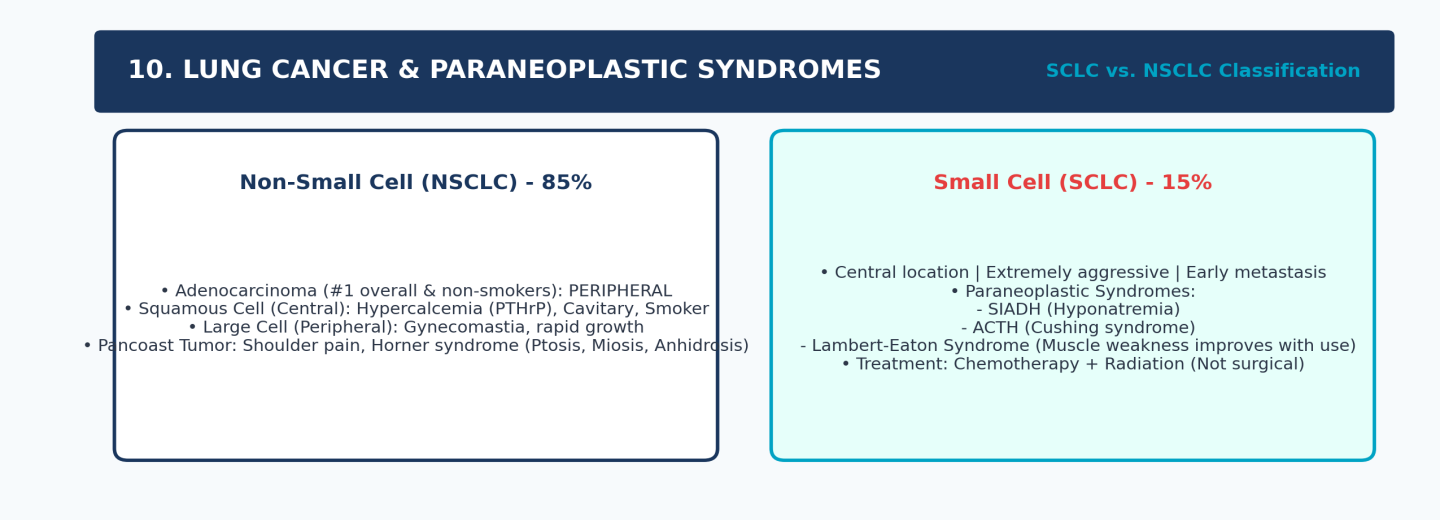
Chest CT followed by tissue biopsy (Bronchoscopy or CT-guided needle biopsy). PET scan for staging.

Pathology Types & Key Paraneoplastics:

- **Adenocarcinoma (40%):** Peripheral | Most common overall & in non-smokers.
- **Squamous Cell (30%):** Central | Cavitary | Smoker | **Hypercalcemia (PTHrP)**.
- **Small Cell (SCLC - 15%):** Central | Aggressive | **SIADH (Hyponatremia), ACTH (Cushing), Lambert-Eaton**.
- **Pancoast Tumor:** Apical lesion -> Horner Syndrome (Ptosis, Miosis, Anhidrosis) + brachial plexus C8-T2 palsy.

MANAGEMENT & TREATMENT

NSCLC: Surgical resection (early stage) +/- Chemotherapy & Radiation. SCLC: Chemotherapy + Radiation (rarely surgical due to early micrometastasis).



HIGH-YIELD BOARD PEARL: BOARD PEARL: Centrally located lung tumor with Hyponatremia = SCLC secreting SIADH! Tumor with Hypercalcemia = Squamous Cell Carcinoma producing PTHrP ('S-Squamous = S-Serum Calcium').

11. PERTUSSIS (WHOOPING COUGH)

PATHOPHYSIOLOGY & ETIOLOGY

Highly contagious upper respiratory tract infection caused by the Gram-negative coccobacillus *Bordetella pertussis*. Pertussis toxin causes respiratory epithelial damage, lymphocytosis, and ciliary paralysis.

CLINICAL PRESENTATION

Classic 3 Clinical Stages:

1. **Catarrhal Stage (Wks 1-2):** Coryza, low-grade fever, mild cough (MOST CONTAGIOUS stage).
2. **Paroxysmal Stage (Wks 2-6):** Paroxysmal fits of severe coughing followed by a high-pitched **inspiratory 'WHOOP'** and post-tussive emesis.
3. **Convalescent Stage (Wks > 6):** Gradual recovery with persistent coughing spells.

PHYSICAL EXAMINATION

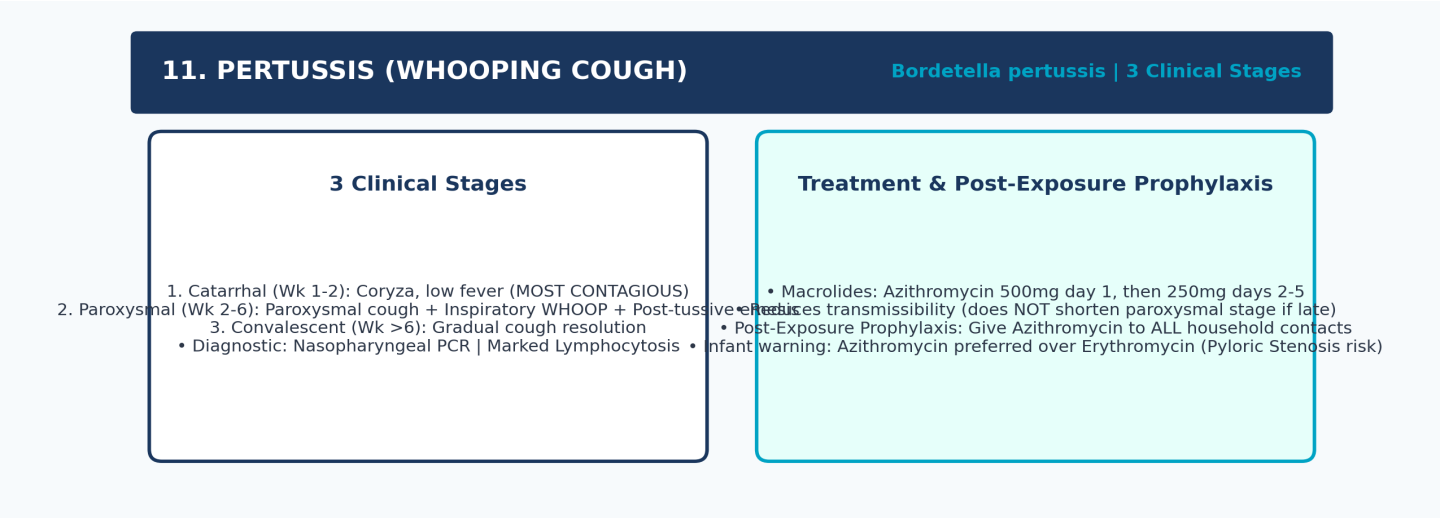
Normal between coughing paroxysms. Petecchiae or subconjunctival hemorrhages secondary to tussive pressure.

DIAGNOSTIC STUDIES

Nasopharyngeal swab PCR or culture (Bordet-Gengou agar). Complete Blood Count reveals **marked Lymphocytosis (WBC > 20,000 with 70% lymphocytes)** despite being a bacterial infection!

MANAGEMENT & TREATMENT

Macrolide Antibiotics: Azithromycin (5-day course) or Erythromycin. Antibiotics eradicate transmissibility but only shorten disease duration if started in Catarrhal stage. **Post-Exposure Prophylaxis:** Azithromycin for ALL close household contacts regardless of vaccination status!



HIGH-YIELD BOARD PEARL: BOARD PEARL: Severe coughing fits with post-tussive emesis + marked LYMPHOCYTOSIS in a bacterial infection = Pertussis. Treat patient AND give Azithromycin prophylaxis to all household contacts!

12. PLEURITIC CHEST PAIN

PATHOPHYSIOLOGY & ETIOLOGY

Sharp, severe, localized chest pain initiated or aggravated by deep inspiration, coughing, or sneezing. Caused by inflammation or irritation of the parietal pleura (innervated by somatic intercostal and phrenic nerves).

CLINICAL PRESENTATION

Sudden onset of sharp, stabbing thoracic pain that limits deep inspiration (catch in breath). Associated dyspnea depending on etiology.

PHYSICAL EXAMINATION

Reproducible chest wall tenderness (if musculoskeletal) or Pleural Friction Rub (scratchy sound during inspiration and expiration).

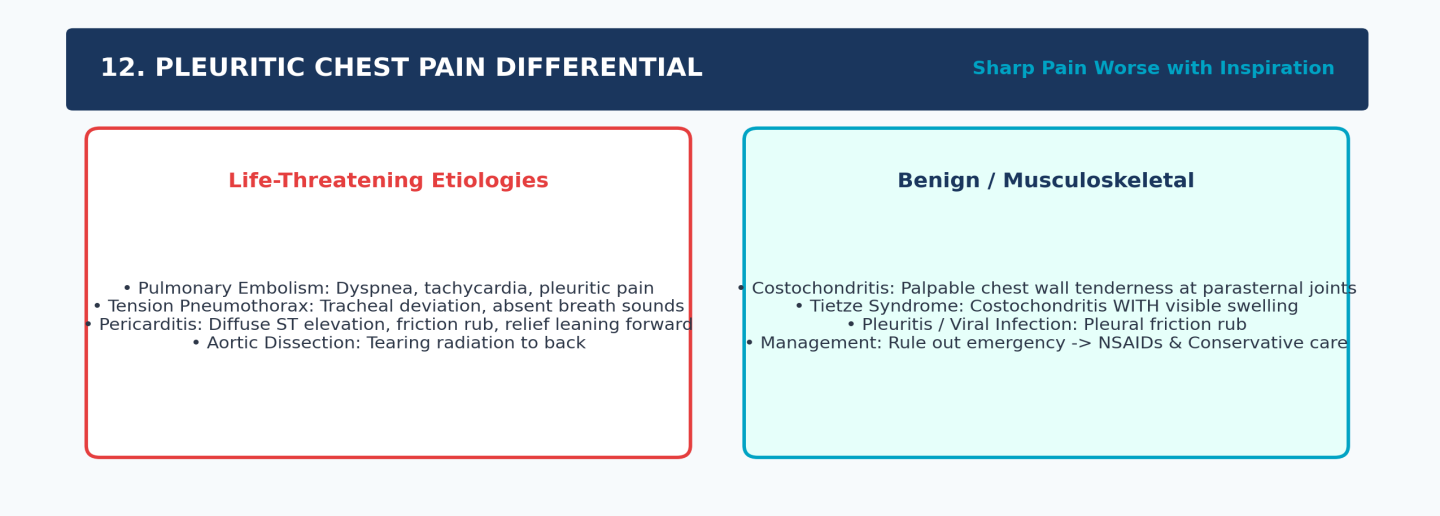
DIAGNOSTIC STUDIES

Immediate diagnostic workup aimed at ruling out life-threatening causes:

- **Pulmonary Embolism:** Dyspnea, tachycardia, PE risk factors -> CTA Chest.
- **Pneumothorax:** Absent breath sounds, hyperresonance -> CXR.
- **Pericarditis:** Relief leaning forward, diffuse ST elevation / PR depression -> ECG.
- **Pneumonia:** Fever, purulent sputum, focal rales -> CXR.
- **Costochondritis:** Reproducible tenderness at costochondral joints.
- **Tietze Syndrome:** Costochondritis WITH visible localized swelling.

MANAGEMENT & TREATMENT

Targeted treatment of underlying emergency. Musculoskeletal costochondritis treated with oral NSAIDs (Ibuprofen/Naproxen) and reassurance.



HIGH-YIELD BOARD PEARL: BOARD PEARL: Point tenderness at the sternal border that reproduces pleuritic pain = Costochondritis. If visible localized joint swelling is present = Tietze Syndrome. Treat both with NSAIDs.

13. PNEUMONIA (CAP, HAP, ATYPICAL & PJP)

PATHOPHYSIOLOGY & ETIOLOGY

Infection of the lung parenchyma leading to alveolar consolidation and inflammatory exudate. Classified as Community-Acquired (CAP), Hospital-Acquired (HAP - >= 48h after admission), or Atypical.

CLINICAL PRESENTATION

Fever, chills, productive cough with purulent or rust-colored sputum, dyspnea, and pleuritic chest pain.

PHYSICAL EXAMINATION

Focal rales/crackles, bronchial breath sounds, increased tactile fremitus, dullness to percussion, egophony ('E' to 'A' change).

DIAGNOSTIC STUDIES

CXR shows lobar consolidation or interstitial infiltrates.

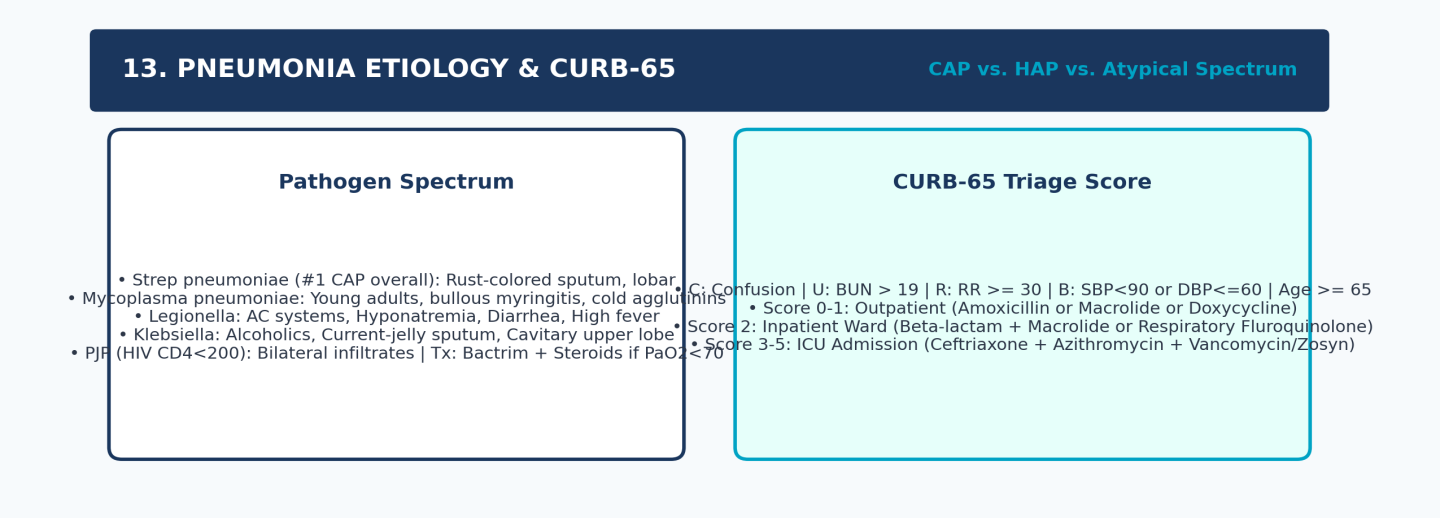
Pathogen Characteristics:

- *Strep pneumoniae:* #1 cause CAP | Rust-colored sputum | Lobar infiltrate.
- *Mycoplasma pneumoniae:* Young adults/dorms | Bullous myringitis | Cold agglutinins.
- *Legionella pneumophila:* AC systems/water | Hyponatremia + Diarrhea + High fever.
- *Klebsiella pneumoniae:* Alcoholics | Current-jelly sputum | Upper lobe cavity.
- *Pneumocystis jirovecii (PJP):* HIV CD4 < 200 | Bilateral diffuse interstitial infiltrates | High LDH.

CURB-65 Risk Score: Confusion, Urea > 19, RR >= 30, BP < 90/60, Age >= 65. (0-1: Outpatient, 2: Inpatient Ward, >= 3: ICU).

MANAGEMENT & TREATMENT

CAP Outpatient: Amoxicillin or Doxycycline or Macrolide. CAP Inpatient: Ceftriaxone + Azithromycin OR Respiratory Fluoroquinolone (Levofloxacin). PJP: Trimethoprim-Sulfamethoxazole (Bactrim) + IV Steroids if PaO2 < 70 mmHg.



HIGH-YIELD BOARD PEARL: BOARD PEARL: Pneumonia in an alcoholic with current-jelly sputum = Klebsiella. Pneumonia in an HIV patient with CD4 < 200 = PJP; add IV Steroids if PaO2 < 70 or A-a gradient >= 35!

14. PULMONARY EMBOLISM (PE)

PATHOPHYSIOLOGY & ETIOLOGY

Occlusion of the pulmonary arterial bed by an embolus, most commonly originating from a Deep Vein Thrombosis (DVT) in the lower extremities (Iliofemoral veins). Leads to ventilation-perfusion (V/Q) mismatch, increased right ventricular strain, and potential cardiogenic collapse.

CLINICAL PRESENTATION

Sudden onset of dyspnea (#1 symptom), pleuritic chest pain, tachypnea, cough, hemoptysis, or syncope.

PHYSICAL EXAMINATION

Tachypnea (RR > 20) and Tachycardia (HR > 100) are most consistent signs. Loud S2, JVD, or signs of DVT (unilateral calf swelling/tenderness).

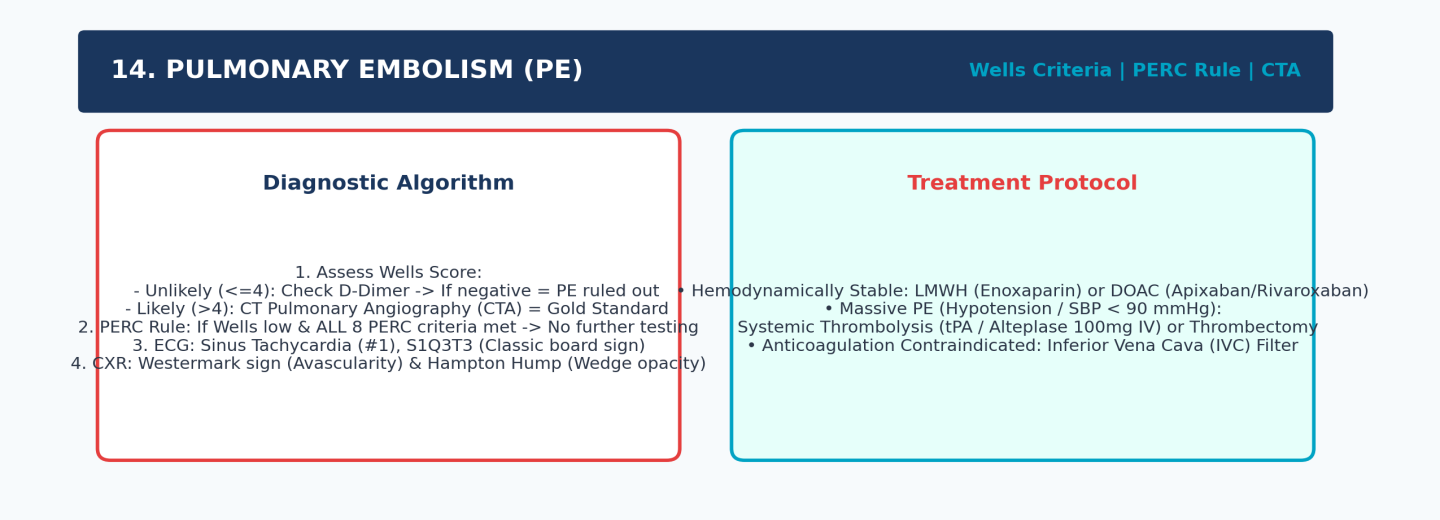
DIAGNOSTIC STUDIES

Wells Criteria Score:

- Unlikely (<= 4): Order High-Sensitivity D-Dimer (If negative, PE ruled out).
- Likely (> 4): **CT Pulmonary Angiography (CTA)** = Initial Diagnostic Test of Choice.
- **PERC Rule:** If low risk & all 8 PERC criteria met -> PE ruled out without testing.
- ECG: Sinus Tachycardia (#1 finding); **S1Q3T3 pattern** (classic board sign of RV strain).
- CXR: Often normal; Hampton's Hump (wedge-shaped opacity) or Westermark Sign.

MANAGEMENT & TREATMENT

Hemodynamically Stable: Anticoagulation with LMWH (Enoxaparin), Fondaparinux, or DOAC (Apixaban/Rivaroxaban).
Massive PE (Hypotension / SBP < 90 mmHg): Systemic Thrombolysis (tPA / Alteplase) or catheter-directed thrombectomy.
Anticoagulation Contraindicated: Inferior Vena Cava (IVC) Filter.



HIGH-YIELD BOARD PEARL: BOARD PEARL: Hypotensive PE patient (Massive PE) requires immediate Systemic Thrombolysis (tPA)! For stable PE, start Anticoagulation before CTA if diagnostic delay is anticipated.

15. RESPIRATORY SYNCYTIAL VIRUS (RSV)

PATHOPHYSIOLOGY & ETIOLOGY

Enveloped, single-stranded RNA virus belonging to the Paramyxoviridae family. Primary cause of lower respiratory tract infections (bronchiolitis and pneumonia) in infants and young children worldwide.

CLINICAL PRESENTATION

Starts with URI symptoms (coryza, low fever) progressing to lower tract involvement: cough, wheezing, tachypnea, intercostal retractions, and poor feeding.

PHYSICAL EXAMINATION

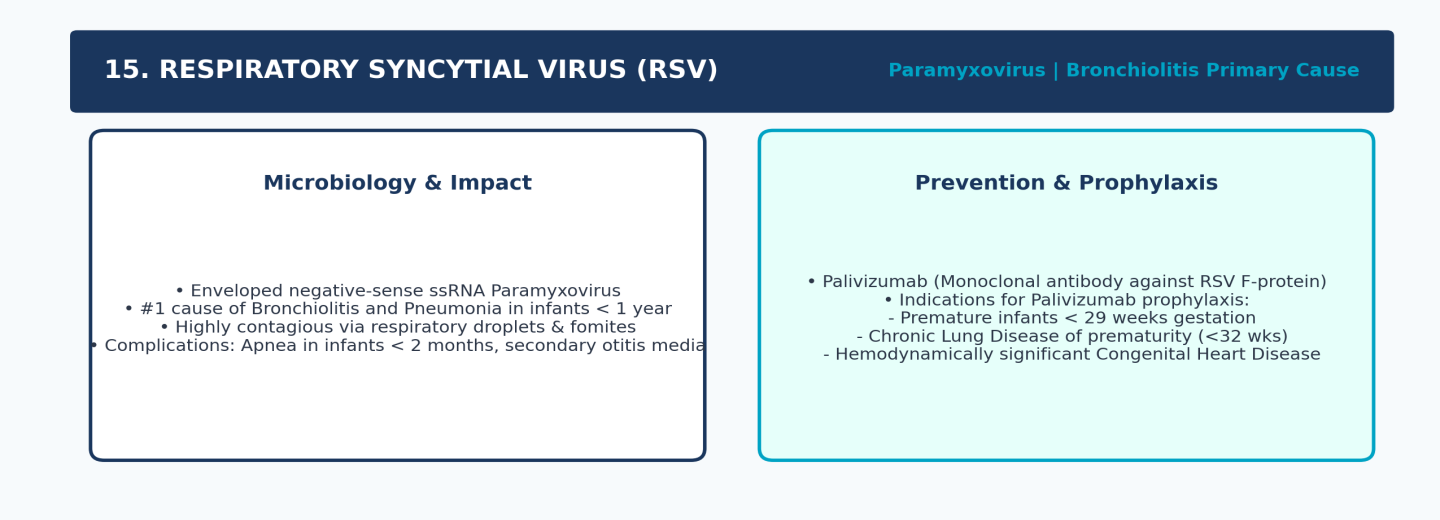
Tachypnea, diffuse expiratory wheezing, rales, nasal flaring. High risk for apnea in infants < 2 months old or premature infants.

DIAGNOSTIC STUDIES

Nasopharyngeal swab RT-PCR or rapid viral antigen test. CXR shows hyperinflation, flattening of diaphragms, and atelectasis.

MANAGEMENT & TREATMENT

Supportive management (nasal suctioning, hydration, oxygen). **Prophylaxis: Palivizumab** (monoclonal antibody targeting RSV F-protein) given monthly during RSV season to high-risk infants (prematurity < 29 weeks, chronic lung disease, or significant congenital heart disease).



HIGH-YIELD BOARD PEARL: BOARD PEARL: RSV is the leading cause of bronchiolitis in infants. Palivizumab prophylaxis is indicated for premature infants < 29 weeks gestation or those with congenital heart disease.

16. SHORTNESS OF BREATH (ACUTE DYSPNEA TRIAGE)

PATHOPHYSIOLOGY & ETIOLOGY

Subjective symptom of breathing discomfort resulting from complex interactions between physiological, psychological, and environmental factors. Acute onset represents a medical emergency until proven otherwise.

CLINICAL PRESENTATION

Sensation of air hunger, chest tightness, or inability to catch breath. May be positional (Orthopnea in HF, Trepopnea in unilateral lung disease).

PHYSICAL EXAMINATION

Red Flags for Immediate Airway/Ventilatory Failure:

- Inability to speak in full sentences | Tripod stance / Accessory muscle use.
- Stridor or Silent Chest | Cyanosis / SpO2 < 90%.
- Altered mental status / Lethargy (Hypercapnia).
- Asymmetric breath sounds (Tension Pneumothorax / Foreign Body).

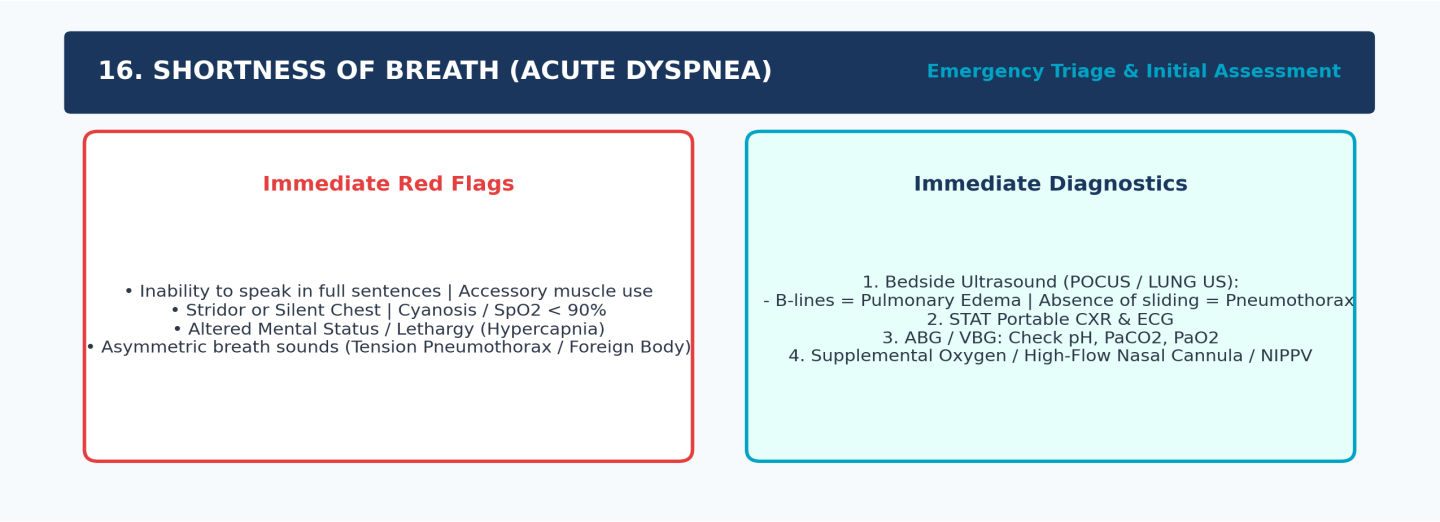
DIAGNOSTIC STUDIES

Immediate Bedside Evaluation:

1. **POCUS (LUNG Ultrasound):** B-lines = Pulmonary Edema | Absence of lung sliding = Pneumothorax.
2. STAT Portable CXR & ECG.
3. ABG / VBG: Assess pH, PaCO2, PaO2.
4. High-sensitivity Troponin, BNP, D-Dimer.

MANAGEMENT & TREATMENT

Immediate oxygenation (Nasal Cannula -> Non-rebreather -> High-Flow Nasal Cannula -> NIPPV / BiPAP -> Endotracheal Intubation). Targeted emergency intervention based on underlying cause.



HIGH-YIELD BOARD PEARL: BOARD PEARL: Bedside Lung Ultrasound (POCUS) quickly differentiates acute dyspnea: Bilateral B-lines = Acute Pulmonary Edema/CHF; Absent lung sliding + barcode sign = Pneumothorax!

17. TUBERCULOSIS (TB)

PATHOPHYSIOLOGY & ETIOLOGY

Chronic airborne granulomatous infection caused by *Mycobacterium tuberculosis* (acid-fast bacillus). Inhaled droplets deposit in alveoli leading to Ghon focus. Can remain latent (LTBI) or reactivate during immunosuppression.

CLINICAL PRESENTATION

Constitutional symptoms: Low-grade evening fever, night sweats, unexplained weight loss, anorexia, and chronic cough (>3 weeks) with hemoptysis.

PHYSICAL EXAMINATION

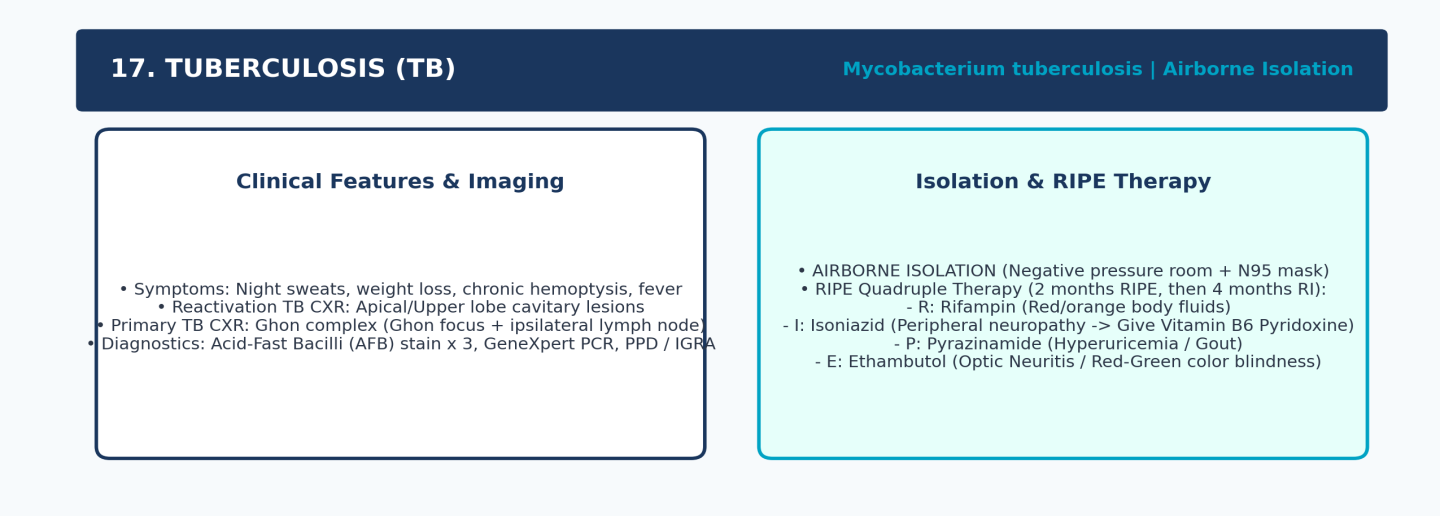
Chronically ill appearance, muscle wasting, post-tussive rales, cachexia.

DIAGNOSTIC STUDIES

• **Chest X-ray:** Reactivation TB shows **Apical / Upper Lobe Cavitary Lesions**; Primary TB shows Ghon complex (calcified primary focus + hilar lymph node).
• **Definitive Diagnosis:** Acid-Fast Bacilli (AFB) sputum smear & culture (Lowenstein-Jensen medium) x 3 consecutive morning samples, or GeneXpert PCR.
• Screening: PPD skin test or IGRA (Quantiferon Gold).

MANAGEMENT & TREATMENT

AIRBORNE ISOLATION (Negative Pressure Room + N95 Mask).
RIPE Quadruple Therapy (2 months RIPE, followed by 4 months RI):
• **R - Rifampin:** Orange/red discoloration of body fluids | Hepatotoxicity.
• **I - Isoniazid:** Peripheral Neuropathy (**Must give Vitamin B6 Pyridoxine!**) | Drug-induced Lupus.
• **P - Pyrazinamide:** Hyperuricemia / Gout | Hepatotoxicity.
• **E - Ethambutol: Optic Neuritis** (decreased visual acuity & red-green color blindness).



HIGH-YIELD BOARD PEARL: BOARD PEARL: Apical cavitary lesions + night sweats = Reactivation TB! Always administer Pyridoxine (Vitamin B6) with Isoniazid to prevent peripheral neuropathy, and monitor for Ethambutol optic neuritis.